

Complex Python

Lesson #1

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Icebreaker!

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Intro to Python



Python is a hybrid scripting language; these types of languages are usually simple and easy to use.

Key characteristics:

- Dynamically typed
- Hybrid
- Extremely Versetile

These traits are part of the reason of why Python is a highly-valued skill in the technological sector.

Downloading Python

To download our starting Python environment, go to python.org and click the Download Python install manager. For the first 3 prompts enter y. For the 4th prompt enter n. The python shell(playground) should appear. Here do can try out simple Python programs.



Datatypes

Common Datatypes:

- Strings(str)
- Integers(int)
- Decimals(float)
- True/False(bool)

Using the type() function is a useful tool in determining a variable's datatype. If the variable does not contain a value then it is marked as None by Python, similar to nil or null in other languages.

Ex.

```
name = "John"
isLocked = True
score = None
print(type(name))
```

If Statements

There are 2 types of operators, comparison(==,!=,>,<, <=,>=) and logical(and, or, not) operators. These can be used in a if statement, if its condition is evaluated to be true. Another form of an if statement is the elif statement, which uses else and elif allowing for multiple condition checking. You can also nest loops, where you put one condition statement inside another condition statement. User input is done through input function.

Ex. grade = int(input("Enter your age: "))

```
if grade >= 90:
    print(“A”)
elif score >= 80:
    print(“B”)
```

Activity Time!

Activity #1

Use the Python Shell to make a variables and test the type of it.

Also create your own if statement with elif and else statements.

An example is shown below:

```
>>> grade = int(input("Enter your grade: "))
... if grade >= 90:
...     print("A")
... elif grade >= 80:
...     print("B")
... elif grade >= 70:
...     print("C")
... else:
...     print("F")
...
Enter your grade: 34
F
```

Review Questions

What is the type() function used for?

It tell us the datatype of the variable

Do I store the float 32.6 in quotes or not?

No, only strings are stored in quotes. Floats do not need them,

Recap

- The main 4 variables are str, int, float, bool. These can be checked using the type() function.
- If statements can be used to check conditions and will lead to an event if the condition is true.

Thank You!

Any Questions?